

## Properties of equality



1 Write the new equation produced for each of the following scenarios:

- |                                                              |                                                               |
|--------------------------------------------------------------|---------------------------------------------------------------|
| <b>a</b> Adding 1 to both sides of $x = 9$                   | <b>b</b> Subtracting 7 from both sides of $x = 14$            |
| <b>c</b> Multiplying 5 to both sides of $x = 10$             | <b>d</b> Dividing 11 to both sides of $x = 99$                |
| <b>e</b> Adding 4 to both sides of $x = -6$                  | <b>f</b> Subtracting 20 from both sides of $x = 2$            |
| <b>g</b> Multiplying 8 to both sides of $x = -3$             | <b>h</b> Dividing 5 to both sides of $x = -25$                |
| <b>i</b> Multiplying 11 to both sides of $5x = 6$            | <b>j</b> Multiplying 12 to both sides of $3x = -8$            |
| <b>k</b> Dividing 3 to both sides of $9x = 27$               | <b>l</b> Dividing 9 to both sides of $45x = -54$              |
| <b>m</b> Multiplying $\frac{7}{8}$ to both sides of $5m = 3$ | <b>n</b> Multiplying $\frac{1}{3}$ to both sides of $-4x = 5$ |

2 Determine whether the following equations are equivalent:

- |                                                         |                                                           |
|---------------------------------------------------------|-----------------------------------------------------------|
| <b>a</b> $-5y = 13$ and $\frac{-5y}{-5} = \frac{13}{5}$ | <b>b</b> $\frac{u}{6} = 4$ and $6 \times \frac{u}{6} = 4$ |
|---------------------------------------------------------|-----------------------------------------------------------|

3 Determine the property that justifies why the equations  $9x = -27$  and  $\frac{9x}{9} = \frac{-27}{9}$  are equivalent.

4 Fill in the blank so that each resulting statement is true.

- a** The addition property of equality states that if  $a = b$ , then  $a + c = \text{ } \mathbb{I}$ .
- b** The multiplication property of equality states that if  $a = b$  and  $c \neq 0$ , then  $ac = \text{ } \mathbb{I}$ .

5 Consider the equation  $3(a + (-6)) + 10(2a + 5) = 8$ . Determine the property that justifies every step.

- a**  $3a + (-18) + 20a + 50 = 8$  is obtained after the first step.
- b**  $-18 + 50 + 3a + 20a = 8$  is obtained after the second step.
- c**  $(-18 + 50) + (3a + 20a) = 8$  is obtained after the third step.

6 A student is in the process of solving an equation. The original equation and the first step are as follows:

$$\text{Original: } 9a + 7 = 2 - 5a + 6$$

$$\text{Step 1: } 9a + 7 = 2 + 6 - 5a$$

Determine the property used by the student for the first step.

7 Consider the following equations:



Equation 1:  $3x = -15$

Equation 2:  $\frac{3x}{3} = \frac{-15}{3}$

- a Determine the property why Equation 1 and Equation 2 are equivalent.
- b State the number that is multiplied to both sides of Equation 1 resulted to Equation 2.

8 Determine whether the following are valid deductions from the statement  $a = 12$ :

- a  $2a = 6$
- b  $a + 7 = 19$
- c  $\frac{a}{2} = 6$
- d  $a + 1 = 12$

9 Determine the property illustrated for each statement.

- a If  $a = d$ , then  $a + 3 = d + 3$ .
- b Given that  $b = 16$ , then  $b - 5 = 11$ .

10 Consider the equation  $21 = x + 13$ .

- a Solve for  $x$ .
- b Kathleen started to solve the equation as follows:

$$\begin{aligned} 21 &= x + 13 \\ 21 - 13 &= x + 13 - 13 \end{aligned}$$

Determine the property that justifies this step of work.

- c Kathleen finished solving the equation as shown:

$$\begin{aligned} 21 - 13 &= x + 13 - 13 \\ 8 &= x + 0 \\ 8 &= x \end{aligned}$$

Determine the property that justifies Kathleen's final step of work.

11 Consider the equation  $7x = 49$ .



- a Solve for  $x$ .
- b Ray started to solve the equation as follows:

$$\begin{aligned}7x &= 49 \\ \frac{7x}{7} &= \frac{49}{7}\end{aligned}$$

Determine the property that justifies this step of work.

- c Ray finished solving the equation as shown:

$$\begin{aligned}\frac{7x}{7} &= \frac{49}{7} \\ 1x &= 7 \\ x &= 7\end{aligned}$$

Determine the property that justifies Ray's final step of work.

12 Consider the equation  $t - 14 = 0$ .

- a Solve for  $t$ .
- b Vanessa started to solve the equation as follows:

$$\begin{aligned}t - 14 &= 0 \\ t - 14 + 14 &= 0 + 14\end{aligned}$$

Determine the property that justifies this step of work.

- c Vanessa finished solving the equation as shown:

$$\begin{aligned}t - 14 + 14 &= 0 + 14 \\ t - 0 &= 0 + 14 \\ t &= 14\end{aligned}$$

Determine the property that justifies Vanessa's final step of work.

- 13 Consider the equation  $\frac{8x + 48}{9} = 16$ .



- a Solve for  $x$ .
- b Dylan started to solve the equation as follows:

$$\frac{8x + 48}{9} = 16$$
$$8x + 48 = 144$$

Determine the property that justifies Dylan's first step of work.

- c Dylan continued to solve the equation as shown:

$$8x + 48 = 144$$
$$8x = 96$$

Determine the property that justifies this step of work.

- d Dylan finished solving the equation as follows:

$$8x = 96$$
$$x = 12$$

Determine the property that justifies Dylan's final step of work.

- 14 Consider the equation  $\frac{2x - 4}{3} = 4$ .

- a Solve for  $x$ .
- b Skye solved the equation as follows:

Given equation:  $\frac{2x - 4}{3} = 4$

Step 1:  $2x - 4 = 12$

Step 2:  $2x = 16$

Step 3:  $1x = 8$

Solution:  $x = 8$

Determine the property that justifies Skye's final step of work.

- 15 Consider the equation  $\frac{4(x+6)}{5} = 20$ .



- a Solve for  $x$ .  
b Susana started solving the equation as follows:

$$\begin{aligned}\frac{4(x+6)}{5} &= 20 \\ 4(x+6) &= 100\end{aligned}$$

Determine the property that justifies Susana's first step of work.

- c Susana continued to solve the equation as shown:

$$\begin{aligned}4(x+6) &= 4 \\ x+6 &= 1\end{aligned}$$

Determine the property that justifies this step of work.

- d Susana finished solving the equation as follows:

$$\begin{aligned}x+6 &= 1 \\ x &= -5\end{aligned}$$

Determine the property that justifies Susana's final step of work.

- 16 Consider the equation  $\frac{2(x-6)}{9} = 6$ .

- a Solve for  $x$ .  
b Ned solved the equation as follows:

$$\begin{aligned}\text{Given equation: } & \frac{2(x-6)}{9} = 6 \\ \text{Step 1: } & \frac{2(x-6)}{9} \times 9 = 6 \times 9 \\ \text{Step 2: } & 2(x-6) = 54 \\ \text{Step 3: } & \frac{2(x-6)}{2} = \frac{54}{2} \\ \text{Step 4: } & x-6 = 27 \\ \text{Step 5: } & x-6+6 = 27+6 \\ \text{Step 6: } & x-0 = 33 \\ \text{Solution: } & x = 33\end{aligned}$$

Determine the property that justifies Ned's final step of work.

17 Consider the equation  $4\left(\frac{x}{6} + 7\right) = 8$ .



- a Solve for  $x$ .
- b Sally started to solve the equation as follows:

$$4\left(\frac{x}{6} + 7\right) = 8$$

$$\frac{x}{6} + 7 = 2$$

Determine the property that justifies Sally's first step of work.

- c Sally continued to solve the equation as shown:

$$\frac{x}{6} + 7 = 2$$

$$\frac{x}{6} = -5$$

Determine the property that justifies this step of work.

- d Sally finished solving the equation as follows:

$$\frac{x}{6} = -5$$

$$x = -30$$

Determine the property that justifies Sally's final step of work.

18 Consider the equation  $6\left(\frac{x}{9} - 4\right) = 18$ .

- a Solve for  $x$ .
- b Irene solved the equation as follows:

Given equation:  $6\left(\frac{x}{9} - 4\right) = 18$

Step 1:  $\frac{6\left(\frac{x}{9} - 4\right)}{6} = \frac{18}{6}$

Step 2:  $\frac{x}{9} - 4 = 3$

Step 3:  $\frac{x}{9} - 4 + 4 = 3 + 4$

Step 4:  $\frac{x}{9} = 7$

Step 5:  $\frac{x}{9} \times 9 = 7 \times 9$

Step 6:  $1x = 63$

Solution:  $x = 63$

Determine the property that justifies Irene's final step of work.